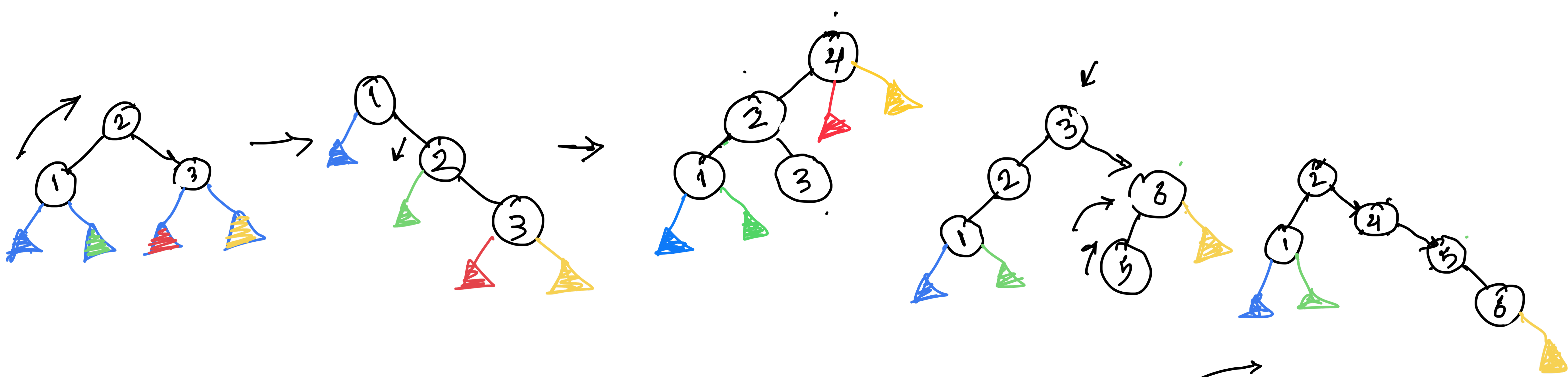
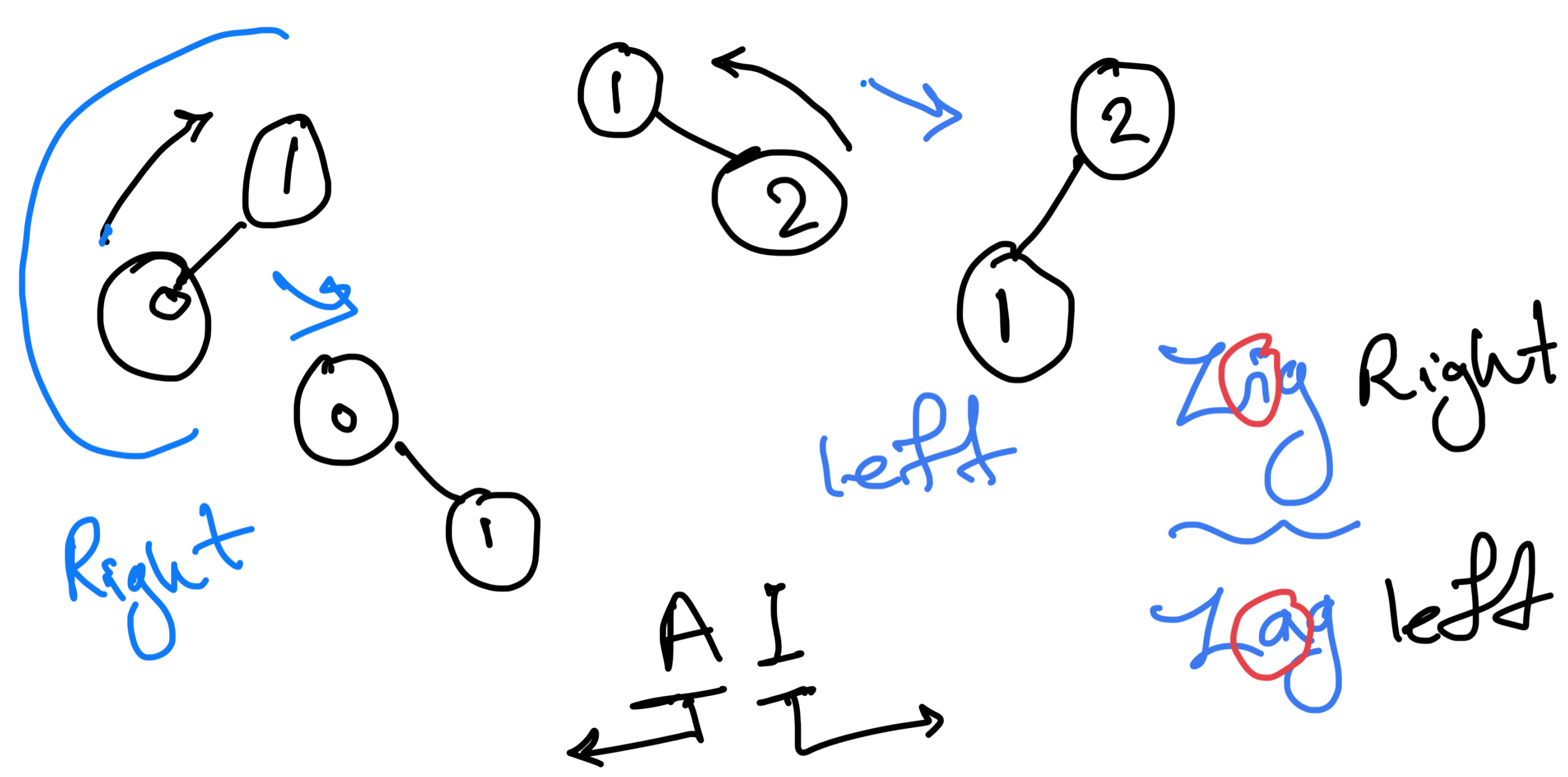




```

58 private void splay(Node x) {
59     while (x.parent != null) {
60         Node p = x.parent;
61         Node g = p.parent;
62         if (g == null) { // Zig
63             if (x == p.left)
64                 rotateRight(p);
65             else
66                 rotateLeft(p);
67         } else if (x == p.left && p == g.left) { // Zig-Zig
68             rotateRight(g);
69             rotateRight(p);
70         } else if (x == p.right && p == g.right) { // Zig-Zig
71             rotateLeft(g);
72             rotateLeft(p);
73         } else if (x == p.right && p == g.left) { // Zig-Zag
74             rotateLeft(p);
75             rotateRight(g);
76         } else { // Zig-Zag
77             rotateRight(p);
78             rotateLeft(g);
79         }
80     }
81 }

```

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